Scorched AI Trading Bot

Improvement Roadmap for Live Trading

From $0.08/day to Institutional Quality

A tiered investment plan for maximizing trading performance

Prepared for:	Scorched AI Trading Bot
Date:	March 28, 2026
Current Capital:	$100,343 ($100K starting)
Current Daily Cost:	$0.08/day ($2.37/month)
Recommendation:	**Tier 2 ($0.50/day / $10 per month)**

Table of Contents

1. Current Performance Summary
2. Cost vs. Impact Analysis
3. Tier 1: Better Decisions ($0.25/day)
4. Tier 2: See More, Know More ($0.50/day)
5. Tier 3: Institutional Quality ($1.00/day)
6. Tier 4: Multi-Agent System ($2.50/day)
7. Tier 5: Institutional Grade (>$5/day)
8. Cumulative Impact Overview
9. Data Source Comparison
10. LLM Architecture Comparison
11. Recommendation

1. Current Performance Summary

After 30 days of live trading with a $100,000 paper portfolio, the bot has produced a modest positive return of +0.34%, with a portfolio value of $100,343. While the unrealized positions are all profitable, the closed trade record reveals structural weaknesses that can be addressed systematically.

Metric	Value	Assessment
30-Day Return	+0.34% ($100,343)	Slightly positive
Closed Trades	12 (5W / 7L)	Below breakeven rate
Win Rate	42%	Needs improvement (target: 55%+)
Avg Win / Avg Loss	+$466 / -$638	Reward/risk ratio: 0.73
Realized P&L	-$2,136	Losses from thesis failures
Unrealized P&L	+$2,479 (5 positions)	All open positions green
Daily Cost	$0.08/day	Extremely cost-efficient
Biggest Losses	FCX (-$1,681), ORCL (-$978)	Thesis identification failures

Current Cost Breakdown

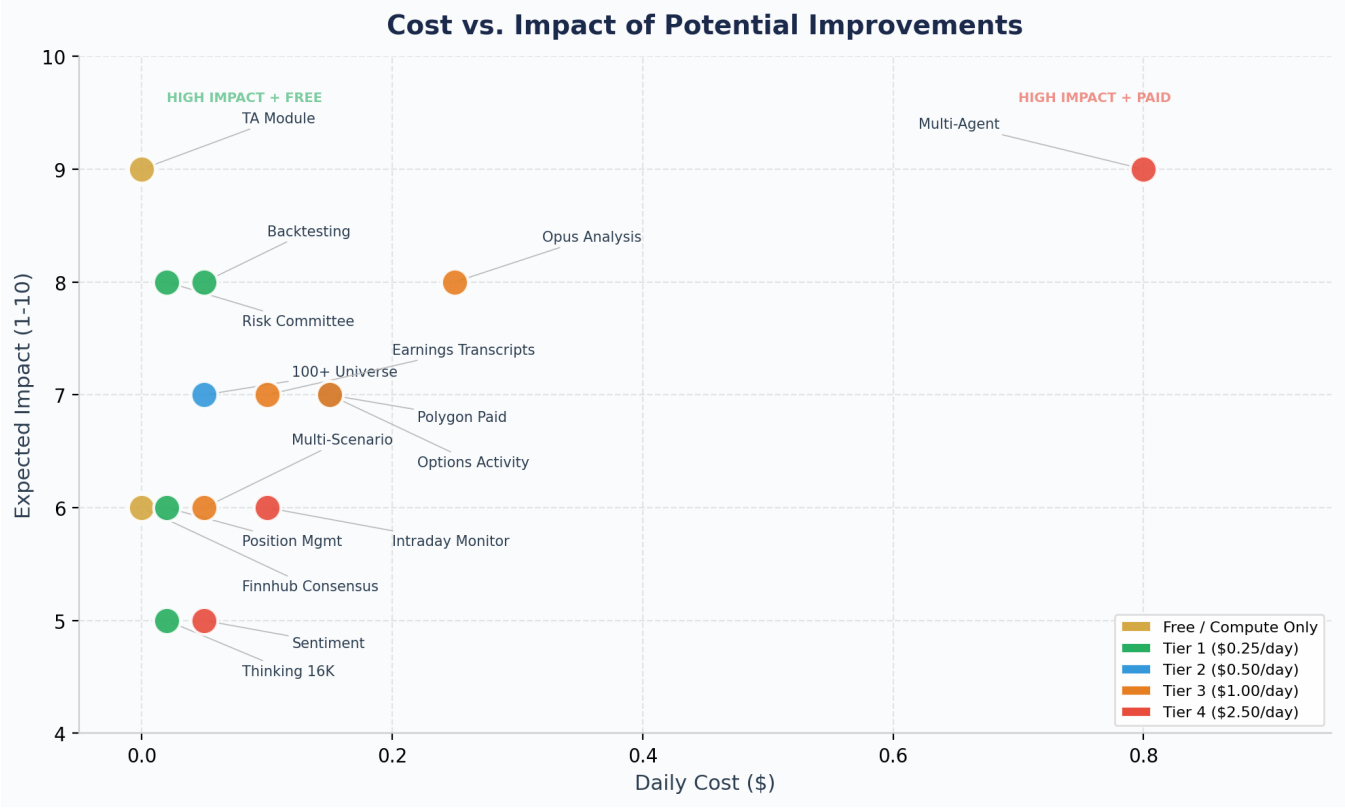
Component	Daily Cost	Monthly
Call 1: Analysis (Sonnet + thinking)	$0.054	$1.62
Call 2: Decision (Sonnet)	$0.012	$0.36
Call 3: EOD Review (Sonnet)	$0.013	$0.39
Total	**$0.079**	**$2.37**

Key Gaps Identified:

- No technical analysis beyond RSI (no MACD, Bollinger Bands, support/resistance)
- No analyst consensus or price targets
- No sentiment data from social media or news
- Single LLM pass with no adversarial review (thesis failures go unchallenged)
- 40-stock watchlist limits opportunity discovery
- Headlines only from news sources (no full article context)
- 15-minute delayed price data via yfinance

2. Cost vs. Impact Analysis

The scatter chart below maps each potential improvement by its daily cost and expected impact on trading performance. The upper-left quadrant represents the highest-value improvements: high impact at zero or minimal cost. Color indicates which tier includes each improvement.



Key Insight: The technical analysis module (MACD, Bollinger Bands, support/resistance) and backtesting framework are the highest-impact improvements at zero marginal cost. They require only compute time, not API spend. The risk committee call at $0.02/day is the single best dollar-for-dollar investment, directly addressing the #1 loss pattern.

3. Tier 1: Better Decisions

$0.25/day ($5/month) | $0.12/day effective

Improvement	Cost	Details
Technical Analysis Module	$0.00/day	Add MACD, Bollinger Bands, and support/resistance levels computed locally. Fills the biggest analytical gap at zero cost. Claude currently makes decisions with only RSI -- like a doctor diagnosing with only temperature.
Risk Committee Call	$0.02/day	A third Claude call that plays devil's advocate, actively trying to poke holes in the trading thesis. Could have prevented both FCX (-$1,681) and ORCL (-$978) by challenging weak catalysts.
Finnhub Analyst Consensus	$0.00/day	Free API providing analyst price targets and consensus ratings. Adds a "Wall Street sanity check" to every recommendation.
Increase Thinking Budget	$0.02/day	Double the extended thinking budget from 8K to 16K tokens. Gives the analysis call more room for deeper reasoning on complex setups.

Key Message: This tier adds a skeptical second opinion and free data sources. The risk committee alone could have prevented the two biggest losses (FCX + ORCL = -$2,659 combined), which exceeds the entire 30-day cost of running this tier for over 5 years.

4. Tier 2: See More, Know More

$0.50/day ($10/month) | $0.35/day effective

Improvement	Cost	Details
Everything in Tier 1	--	All Tier 1 improvements carry forward.
Expand Universe to 100+ Stocks	$0.05/day	Widen the momentum screener beyond the current 40-stock watchlist. The bot's best trades (HAL, CF, SLB) came from the screener, not the watchlist.
Polygon Paid Tier	$0.15/day	Full article text instead of headlines only. Claude currently guesses article content from titles -- like reading a book by its cover.
Position Management Call	$0.02/day	An EOD call focused on open positions: should stops be tightened? Has the thesis changed? Proactive instead of reactive.

Key Message: A wider opportunity set combined with proactive position management. The best trades consistently came from the screener discovering stocks the watchlist missed. Full article text transforms Claude's news analysis from headline speculation to informed assessment.

5. Tier 3: Institutional Quality Analysis

$1.00/day ($20/month) | $0.70/day effective

Improvement	Cost	Details
Everything in Tier 2	--	All Tier 1 + Tier 2 improvements carry forward.
Upgrade Call 1 to Opus	$0.25/day	Opus-quality reasoning for the analysis call catches thesis errors that Sonnet misses. The analysis call is where conviction is formed -- it deserves the strongest model.
Earnings Call Transcripts	$0.10/day	Management commentary from recent earnings calls. Leading indicator of fundamental direction that price data alone cannot capture.
Multi-Scenario Simulation	$0.05/day	Run bull/bear/base scenarios for each candidate. Forces explicit consideration of downside risk before entering positions.
Unusual Options Activity	$0.15/day	Smart money flow data. Large unusual options trades often precede significant price moves by 1-5 days -- exactly the bot's holding period.

Key Message: Opus-quality reasoning catches thesis errors that Sonnet misses. Earnings transcripts and options flow are leading indicators -- they tell you what is about to happen, not what already happened. This tier transforms the bot from reactive to predictive.

6. Tier 4: Multi-Agent System

$2.50/day ($50/month) | $1.50/day effective

Improvement	Cost	Details
Everything in Tier 3	--	All Tier 1-3 improvements carry forward.
Full Multi-Agent Pipeline	$0.80/day	Three specialized agents: Macro Analyst (rates, sector rotation, risk-on/off), Stock Analyst (fundamentals, technicals, catalysts), Portfolio Manager (sizing, correlation, risk budget). Each excels at their domain.
Intraday Monitoring	$0.10/day	11 AM and 2 PM check-in calls. Prevents holding through flash crashes or sudden reversals. Enables same-day exits when thesis breaks.
Sentiment Analysis	$0.05/day	Social media and news sentiment scoring. Identifies crowded trades and contrarian opportunities.
Backtesting Framework	$0.05/day (amortized)	Run strategy variants against historical data. Validates ideas before risking real capital. The only way to distinguish skill from luck.

Key Message: Specialized agents each excel at their domain rather than one generalist handling everything. Intraday monitoring prevents holding through crashes. The backtesting framework provides the feedback loop needed to continuously improve strategy parameters.

7. Tier 5: Institutional Grade

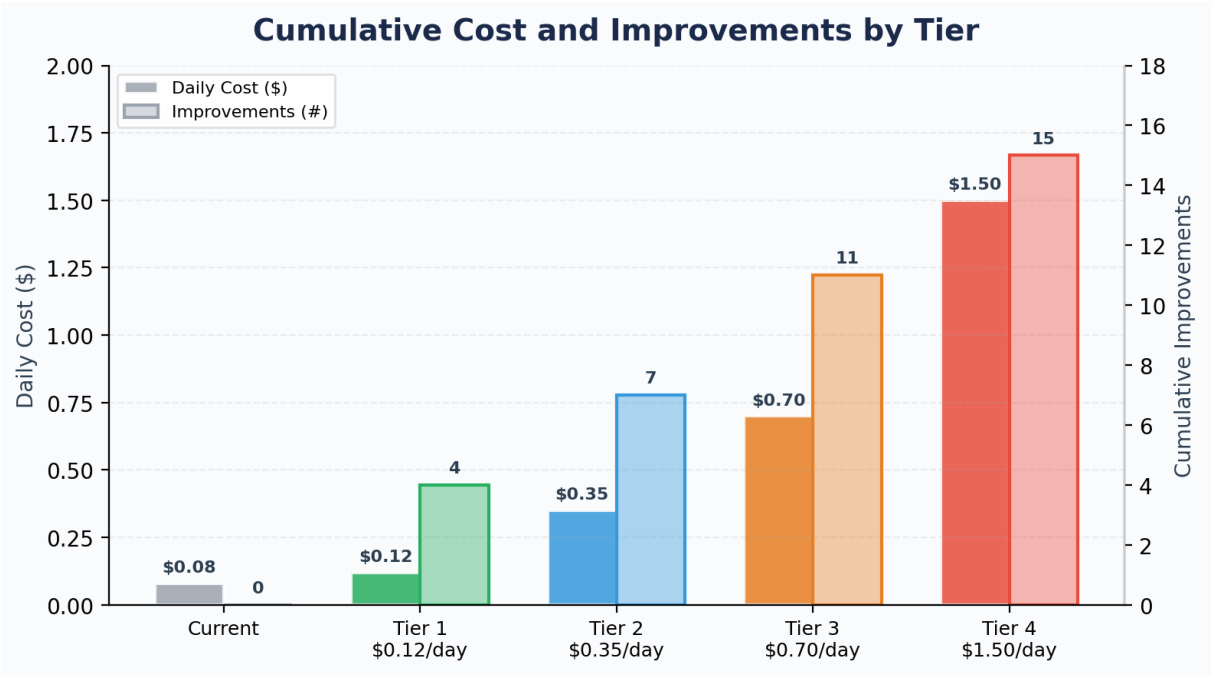
>$5/day ($150+/month) | Variable

Improvement	Cost	Details
Opus Everywhere	$2-5/day	Use Opus for all calls including decision, risk review, and position management.
Bloomberg Terminal API	$65/day	The gold standard for financial data. Real-time, comprehensive, and expensive. Only makes sense at $50K+ capital.
Alternative Data	$2-15/day	Satellite imagery, credit card data, web traffic analytics. Genuine information edge, but expensive.
Full Russell 3000 Universe	$1-3/day	Screen the entire US equity market. Maximum opportunity discovery at maximum compute cost.

Key Message: This tier only makes economic sense at $50,000+ in capital. Bloomberg alone costs $65/day -- 24 times the value of a $1,000 trading account annually. File under "aspirational" until the portfolio justifies it.

8. Cumulative Impact Overview

Each tier builds upon the previous, creating a compounding effect. The chart below shows how daily cost and the number of active improvements scale together across tiers.



Tier	Daily Cost	Monthly	Annual	% of $100K Capital
Current	$0.08	$2.37	$29	0.03%
Tier 1	$0.12	$3.60	$44	0.04%
Tier 2	$0.35	$10.50	$128	0.13%
Tier 3	$0.70	$21.00	$256	0.26%
Tier 4	$1.50	$45.00	$548	0.55%

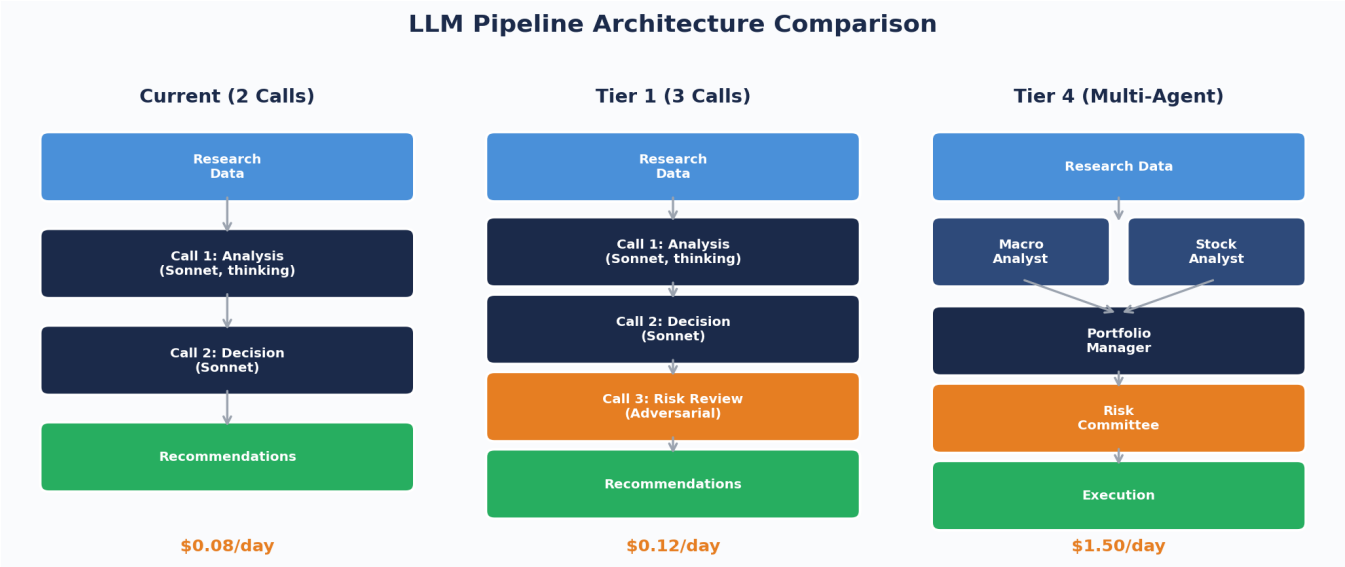
9. Data Source Comparison

The table below shows what data Claude has access to at each tier. Each tier is cumulative -- all capabilities from lower tiers carry forward.

	Current	Tier 1	Tier 2	Tier 3	Tier 4
Price Data (delayed)	YES	YES	YES	YES	YES
Fundamentals	YES	YES	YES	YES	YES
News Headlines	YES	YES	YES	YES	YES
Full Article Text	--	--	YES	YES	YES
Macro Indicators (FRED)	YES	YES	YES	YES	YES
RSI (14)	YES	YES	YES	YES	YES
Full Technicals (MACD/BB/S-R)	--	YES	YES	YES	YES
Analyst Consensus	--	YES	YES	YES	YES
Earnings Transcripts	--	--	--	YES	YES
Unusual Options Flow	--	--	--	YES	YES
Sentiment (Social/News)	--	--	--	--	YES
Real-Time Data	--	--	--	--	--

10. LLM Architecture Comparison

The current two-call pipeline handles everything in a single analysis pass with no adversarial review. Tier 1 adds a risk committee call that challenges the thesis before execution. Tier 4 implements a full multi-agent system with specialized roles.



	Current	Tier 1	Tier 4
Claude Calls/Day	3	4	8-10
Models Used	Sonnet only	Sonnet only	Opus + Sonnet
Adversarial Review	None	Risk Committee	Full Committee
Specialization	Generalist	Generalist + Skeptic	3 Specialists
Position Monitoring	EOD only	EOD only	Intraday (3x)
Estimated Daily Cost	$0.08	$0.12	$1.50

11. Recommendation

RECOMMENDED: Tier 2

$0.50/day | $10/month | $128/year

Why Tier 2?

- **Technical analysis module is FREE** and fills the single biggest analytical gap. MACD, Bollinger Bands, and support/resistance levels are computed locally with zero API cost.
- **Risk committee prevents thesis failures** -- the #1 loss pattern. An adversarial third call at $0.02/day could have saved $2,659 in the first month alone.
- **Expanded universe catches more opportunities.** The bot's best trades (HAL +$891, CF +$445, SLB +$367) all came from the momentum screener, not the static watchlist.
- **Polygon full articles give Claude real context** instead of guessing article content from headlines. At $0.15/day, this transforms news analysis from speculation to assessment.
- **Analyst consensus adds a Wall Street sanity check** at zero cost via Finnhub free tier.

Return on Investment:

Annual cost of Tier 2: **$128**
Cost as % of $100K capital: **0.13%**
FCX + ORCL losses (preventable): **$2,659**
Breakeven: Tier 2 pays for itself if it prevents **one bad trade per year**.

At scale ($50K+ capital), even Tier 3 at $256/year represents just 0.5% of capital -- well below the 1-2% expense ratio of most actively managed funds.

Suggested Implementation Timeline

Week	Action	Expected Result
Week 1	Implement technical analysis module (free)	MACD, BB, S/R levels in every analysis
Week 2	Add risk committee call + Finnhub consensus	Adversarial review catches weak theses
Week 3	Expand universe + add position management	Wider opportunity set, tighter stop management

Week 4	Upgrade to Polygon paid tier	Full article text improves news analysis
Week 5+	Monitor results and evaluate Tier 3	Data-driven decision on further investment

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